

CS 601: CS Research Methodology Seminar

Chapter 2: Goals of Ph.D. Research

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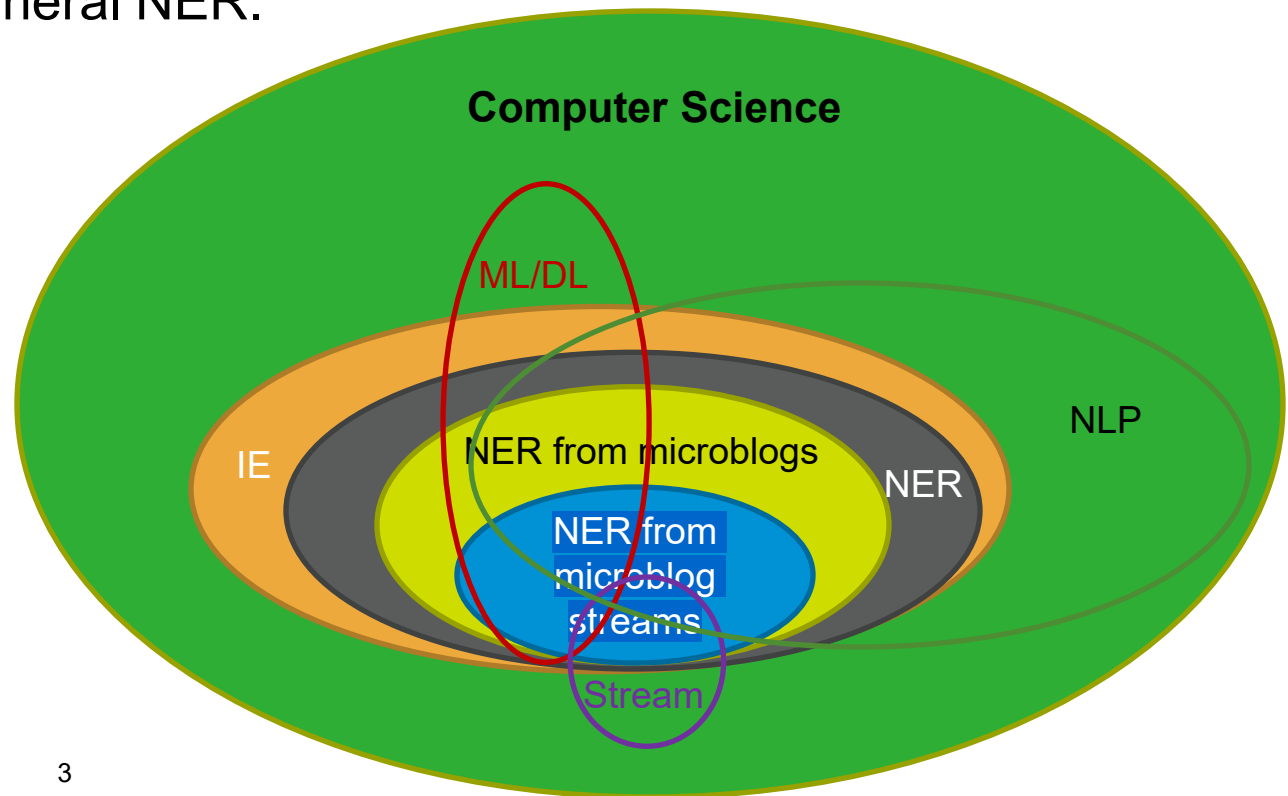
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Goal #1: Be the Best in the Field

- PhD students should aim to become the best expert, at least one of the best experts, in an important, albeit narrow field (i.e., the dissertation topic) and their research should be among the best in the world (in terms of novelty and significance) in the said field by the time when they complete their PhD study. They should also be a top expert in the broader area of the said field.
- To achieve this goal, you need to
 - Be familiar with the literature in the broader area, from classical work to the most recent significant publications, and to industrial trend.
 - Be super familiar with the literature in the focused dissertation topic.
 - Publish multiple papers in the top conferences or journals in the area of the dissertation. In CS, conferences are more important venues than journals for publications. (Csrankings can be a reference point for top conferences in different field.)

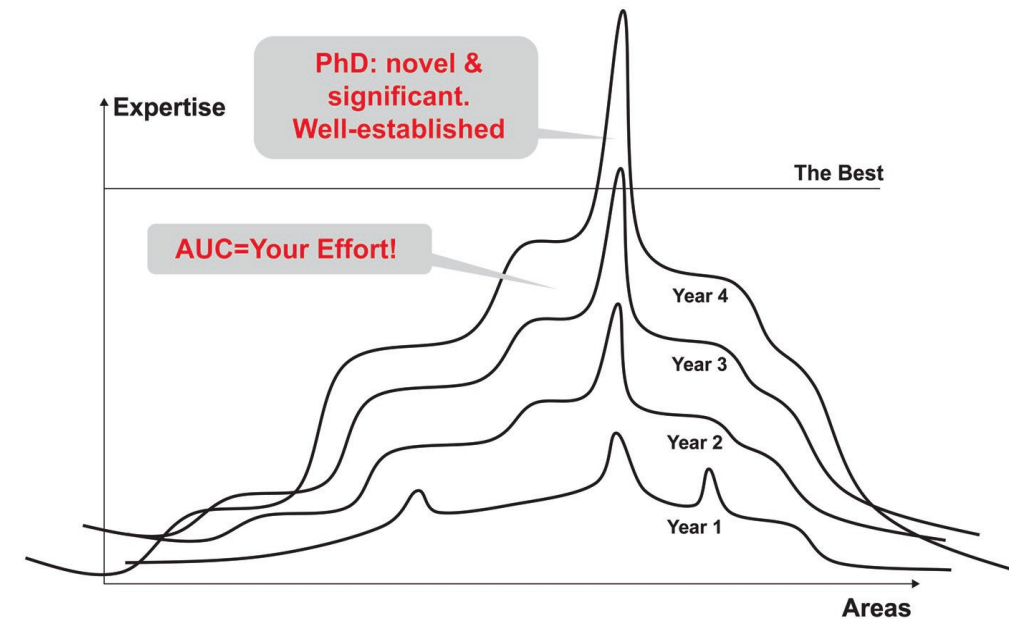
Goal #1: Be the Best in the Field (continued)

- Example. A recent PhD graduate's dissertation topic was on NER (Named Entity Recognition) from microblog streams.
 - She published several high quality papers in this area (two ICDE papers and one TKDE paper).
 - She must be super familiar with papers in this specific topic area but also in a slightly larger area of NER from microblogs and even general NER.
 - She was also familiar with the literature in the broader area of IE (Information Extraction), NLP (Natural Language Processing) and streaming data processing.
 - Since she applied machine learning and deep learning techniques in her research, she also became familiar with ML/DL techniques.



Goal #1: Be the Best in the Field (continued)

- An example scenario of research progress towards a quality PhD dissertation.
 - Year 1. Read papers about a small number of possible dissertation topics and settle on one of them.
 - Year 2. Become more familiar with the literature on the chosen topic and make some progress towards a paper submission.
 - Year 3. Publish the first paper(s), attend and present at 1-2 conferences.
 - The first first-author paper usually is the most difficult.
 - Year 4. Publish additional papers, establish as a true expert in the field of the dissertation topic, write and defend the dissertation.
- PhD in CS usually takes 4-6 years to complete.

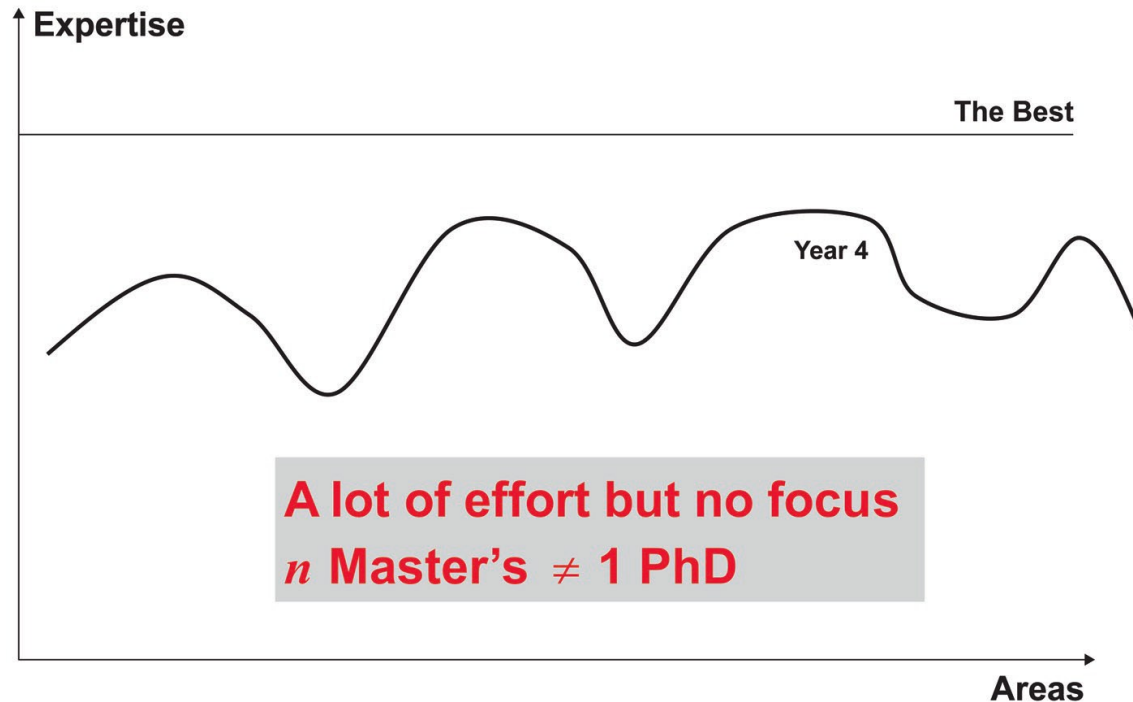


Progress in four years of Ph.D. study.

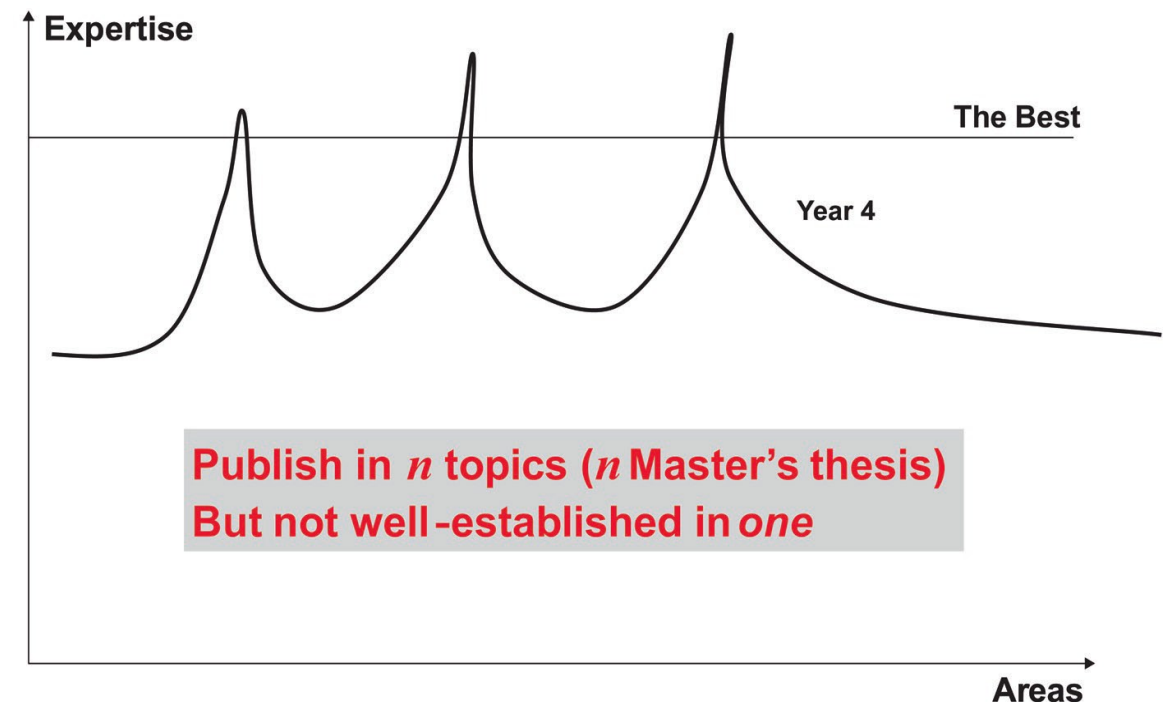
Goal #1: Be the Best in the Field (continued)

- Some examples of poor dissertations

A Poor PhD Thesis...



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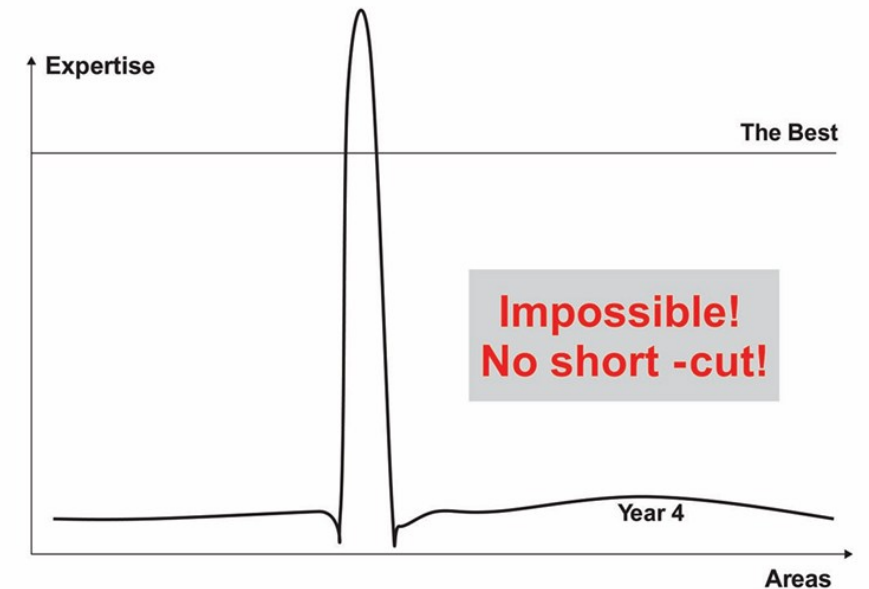


Goal #1: Be the Best in the Field (continued)

- A few exceptionally strong and productive Ph.D. candidates do manage to advance research in more than one area.
- But their Ph.D. thesis usually only consists of published papers in one area.
- A Ph.D. thesis should be coherent, and focus on one (relatively broad) topic.
- Of course, when you become a university professor after your Ph.D., you can develop several strong research areas, working with Ph.D. candidates who have different interests.
- In the third situation, some Ph.D. candidates are keen to find some research topics that can be advanced with little effort (low AUC). They wish they could just focus on a very narrow problem.

Goal #1: Be the Best in the Field (continued)

But in this case, you are not learning to be an independent researcher, your contribution is usually minor, and you do not own the area.



Goal #2: Be Independent

- PhD students should aim to *become an independent researcher* so that you can do tasks listed in Chapter 1, especially Tasks 1–3 (briefly, find new ideas, do solid research, and publish papers in top venues), well by themselves.
- After a student completes his/her PhD study, if they want to continue to do PhD level work, such independence is a must. In fact, they may also need to do other tasks in Chapter 1 on their own.
- The sooner a PhD student becomes independent, the better (see figure).
- This requires the student to develop the skills and abilities crucial for researchers (see Chapter 1).



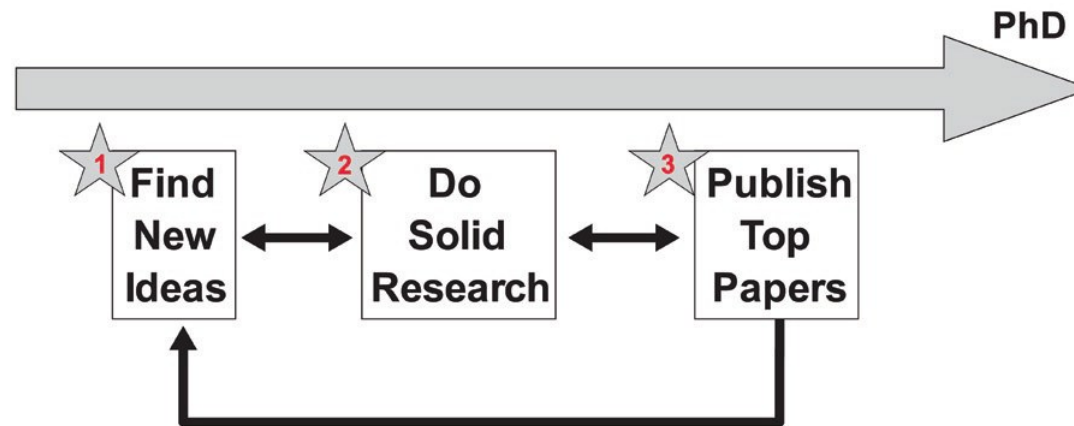
Goal #2: Be Independent

- More specifically, in the first two years, we (i.e. supervisors) usually hold daily or weekly meetings with them discussing new ideas, methods, and experimental results in detail.
- In the later years, they usually met weekly, or whenever needed, to discuss ideas, approaches, and results at high-level. They can extend and modify the ideas, design and modify experimental settings, and write up the paper drafts, mostly by themselves.



Three key tasks in getting a PhD degree

- After the dissertation topic has been decided, the three key tasks for completing a PhD degree are



- This process may need to be repeated multiple times during PhD study.
- The three key tasks will be discussed in detail later.
- With multiple quality papers published, they can be integrated into a PhD dissertation.

Three key tasks in getting a PhD degree

- Through several iterations, you broaden and deepen your Ph.D. topic by building up enough research results and publishing enough papers.
- A good rule of thumb is that you have published a good number of papers in reputable journals and conferences based on your Ph.D. topic.
- Note that although your main job as a Ph.D. candidate is Tasks 1–3, often you also work with your supervisor on all other tasks
- For Example, Helping in grant writing, contributing to outreach events and sometimes with teaching.
- By the end of your Ph.D., you should have made significant contributions in your Ph.D. thesis, and established yourself as a young, independent, and competent researcher!

Distractions and Barriers to the Goals

- Culture shock for new international students.
- Language barrier for students whose native language is not English.
- Being a TA may take away up to 20 hours a week from your research.
- Some have families to take care of or other family-related challenges.
- Not able to make progress with research.
- Losing confidence after repeated rejections of paper submissions.
- Temptation to get a full-time job before completing dissertation research.
- Don't have a balanced life between research and other aspects of life needed to keep it up for 4-6 years. Need to have fun while working hard.
- Don't find time to do exercise and physical activities regularly.
- Poor time management skill.
- Which ones relate to you?
- Anything else?

Some Suggestions

- **Balance:** You have to learn to balance different parts of a graduate-student life: teaching and researching, among others
- Perform their TA duties effectively, and ensure a sufficient amount of time to be reserved for research.
- **Create a balance:** This need to balance one's life is the same if one has to raise a family, learn how to drive, travel for fun, participate in sports and social activities, and generally, enjoy one's life.
- **Health is first:** Even with a high demand in research, Ph.D. students (and faculty members) must find time to do physical exercise to keep themselves fit and healthy.
- **Time Management:** Allocate time for each task and block hours in your calendar for both fun and work, and stick to it.
- Some Ph.D. candidates do work hard, and enjoy doing research, to the point that they do not care how long it takes to get a Ph.D. **Try to enjoy the pain.**